

Worked Answer Key

1. **P1:** 1) Add 5: 27, 32. 2) Subtract 5: 25, 20. 3) Add 10: 146, 156.
2. **P2:** 1) Add 7: 31, 38. 2) Subtract 8: 60, 52. 3) Add 4: 5, 9, 13, 17, 21 chairs.
3. **P3:** 1) Multiply by 3: 162, 486. 2) Divide by 2: 20, 10. 3) Multiply by 4: 256, 1024.
4. **P4:** 1) $3(7) + 2 = 23$. 2) Add 3: output 23. 3) Multiply by 3: $3(10) = 30$.
5. **P5:** 1) $y = 9, 13, 20$. 2) $y = 2, 6, 14$, so $(1, 2), (3, 6), (7, 14)$. 3) Add 9: $y = 34$.
6. **P6:** 1) $y = 2x + 4$. 2) $y = \frac{5}{2}x$ (or multiply by 2.5). 3) $y = 5x + 12$.
7. **P7:** 1) $9 + 14 = 23$. 2) $4(7) = 28$. 3) $3(6) + 2 = 20$.
8. **P8:** 1) $q + 8$. 2) $5r$. 3) $t - 6$.
9. **P9:** 1) Let b be pencils per box: $7b$. 2) Let d be dollars: $d + 12$. 3) Let p be read: $40 - p$.
10. **P10:** 1) $x = 43 - 18 = 25$. 2) $a = 60 - 27 = 33$. 3) $y = 52 - 15 = 37$.
11. **P11:** 1) $z = 29 + 16 = 45$. 2) $c = 75 - 38 = 37$. 3) $h = 19 + 24 = 43$.
12. **P12:** 1) $x = 45 \div 5 = 9$. 2) $n = 63 \div 7 = 9$. 3) $p = 28 \div 4 = 7$.
13. **P13:** 1) $q = 8(6) = 48$. 2) $r = 72 \div 9 = 8$. 3) $s = 11(5) = 55$.
14. **P14:** 1) $n + 35 = 82$, $n = 47$. 2) $6a = 54$, $a = 9$ apples. 3) $s - 18 = 27$, $s = 45$ stickers.
15. **P15:** 1) Terms: 6, 15, 24, 33, 42, 51; term 6 is 51. 2) 3, 6, 12, 24, 48, 96; terms 5 and 6 are 48 and 96. 3) $12 - 3(7) = -9$; term 8 is -9.
16. **P16:** 1) Add 7; term 5 is 39. 2) Multiply by 3; next is 108. 3) Missing 35; subtract 5.
17. **P17:** 1) $A = 18$, $B = 16$; A is greater. 2) $A = 15$, $B = 17$; B is greater. 3) B is always greater by 4 because both share $4x$.
18. **P18:** 1) Rule $8 + 2(n - 1)$; row 7 has 20 seats. 2) $14 + 3(6) = 32$ cm. 3) $4c = 20$; $c = 5$ dollars.
19. **P19:** 1) Multiply by 3: 135, 405. 2) $2(9) + 7 = 25$. 3) $w = 71 - 26 = 45$.
20. **P20:** 1) $a_n = 9 + 6(n - 1)$; term 10 is $9 + 54 = 63$. 2) $y = 4, 10, 19$. 3) $2m + 5 = 23$, so $m = 9$ cards.